# Bank Marketing

## Subscription Prediction

**Aaron Darcy | Abey Mathew**

MSc Data Science - Machine Learning

# Problem

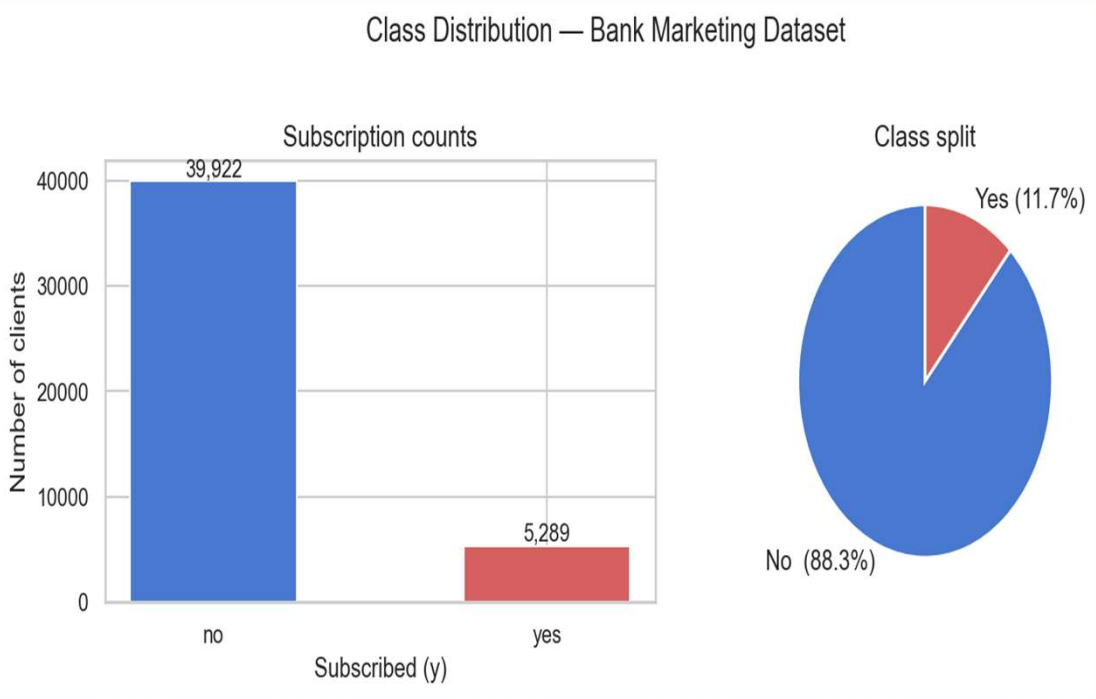
A bank ran a phone campaign selling term deposits. Most people said no.

A term deposit is a savings product where a customer locks money away for a fixed period in return for a guaranteed interest rate. Banks sell these through phone campaigns.

The bank contacted over 45,000 clients. The problem is that only 11.7% said yes. That is a very imbalanced result and it creates a real challenge for machine learning.

A model that simply predicts no for every single person would be 88% accurate. That sounds good but it is completely useless to the bank because it never identifies a single subscriber.

**Our goal: build a model that can find the 12% who will say yes, so the bank can focus its calls on the right people.**




| 45,211 | 11.7% | 88.3% |
|---|---|---|
| contacts made | subscribed | said no |

# Data

UCI Bank Marketing dataset: one row per client contact, 16 features per row

The dataset comes from the UCI Machine Learning Repository. Each row represents one phone call from a bank campaign. The features describe three things:

### Who the client is

Age, job, marital status, education, account balance and whether they have existing loans.

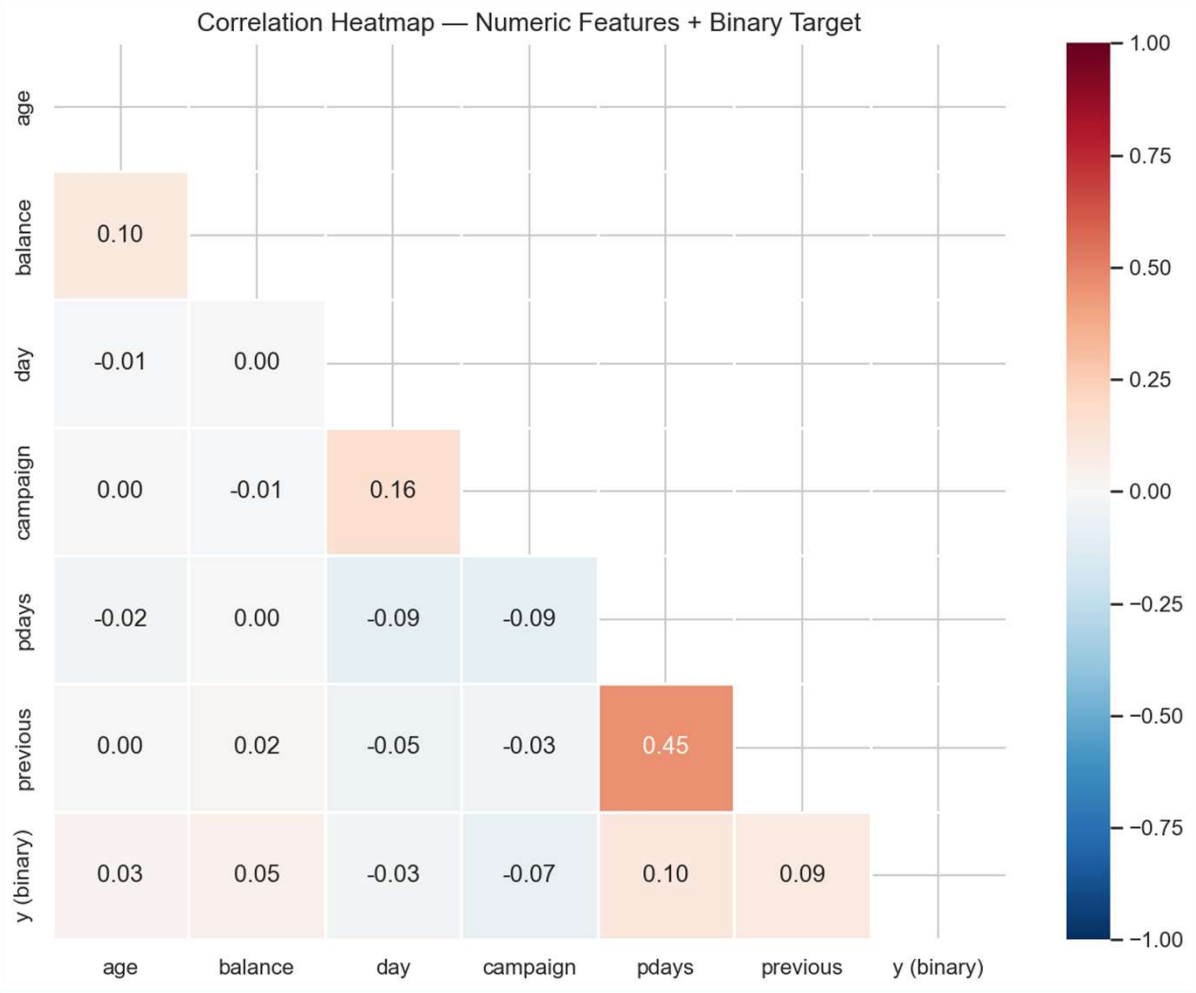
### How the campaign contact went

Number of calls made this campaign, contact method, the month and day of the call.

### Prior campaign history

Whether the client was contacted in a previous campaign, how many days ago, and what the outcome was.

One feature called duration records the call length in seconds. We removed it immediately: it is only known after the call ends and correlates strongly with the outcome for the wrong reason.



# Cleaning

Four decisions made before any model saw the data

Raw data is rarely ready for a machine learning model. These four steps ensured the data was fair, consistent and free from information leakage.

1

### Remove duration

Call length in seconds correlates 0.39 with the outcome but is only known after the call ends. Keeping it would let the model use future information to make predictions. This is called data leakage.

2

### Fix unknowns selectively

job (0.6% unknown) and education (4.1% unknown) are genuinely missing and were replaced with the most common value. contact and poutcome unknowns were kept as valid categories: they carry real meaning.

3

### Handle the pdays sentinel value

81.7% of rows had pdays = -1, which means the client was never contacted before. This is not a number of days. We replaced it with 0 and added a new binary column called contacted_before.

4

### Encode and split

All text columns are converted to 0 and 1 columns using one-hot encoding. The data is then split 80/20 into training and test sets using stratified sampling so both sets have the same 11.7% positive rate.

# Patterns

Exploratory analysis before modelling: what does the data tell us?

Before training any model, we explored the data to understand which features are most associated with subscription. Three patterns stand out clearly.

### Prior outcome is everything

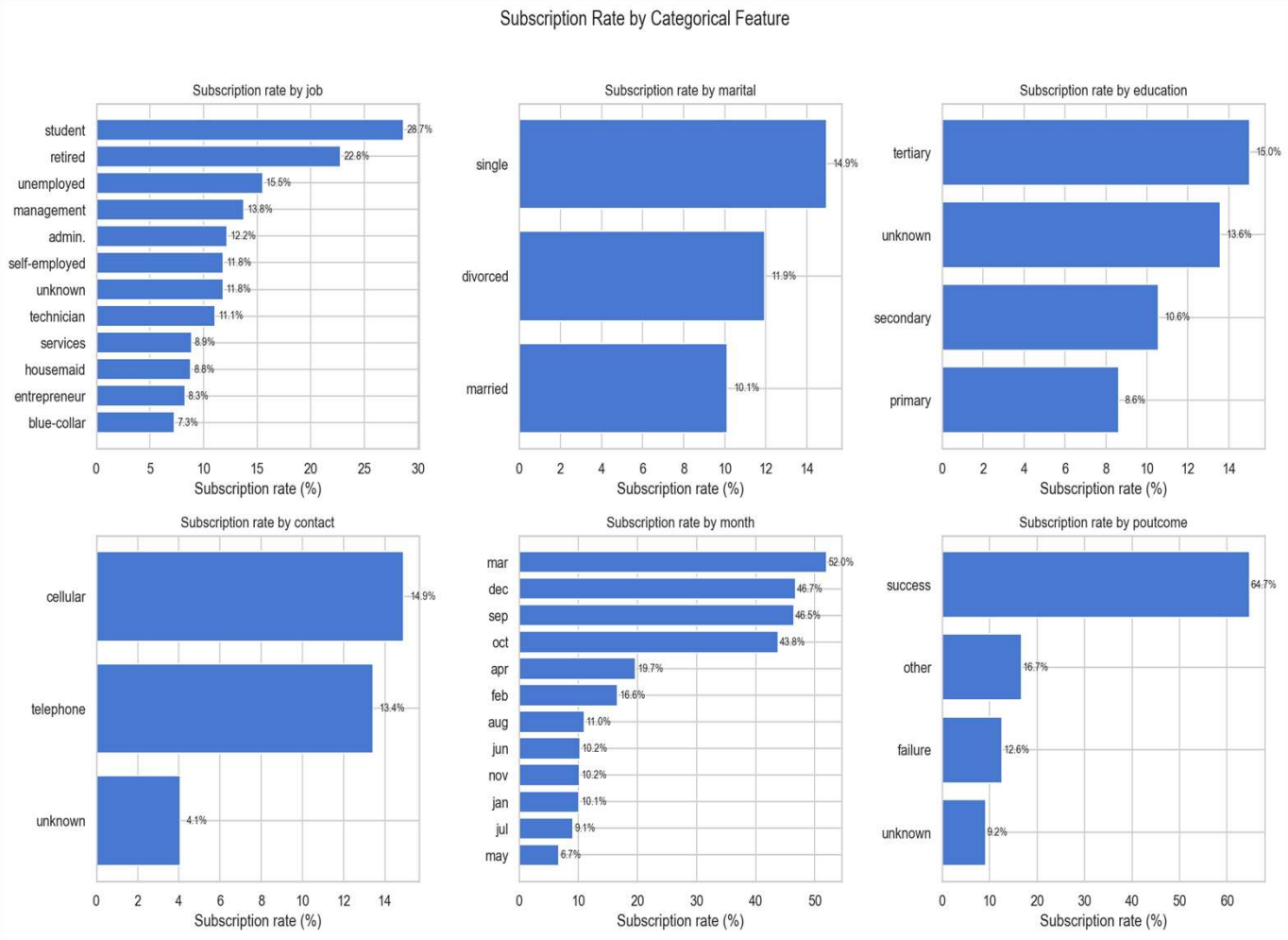
Clients whose previous campaign ended in success convert at 64.7% this time. Clients never contacted before convert at just 9.2%.

### Campaign timing matters more than volume

May has the highest number of calls in the dataset but only 6.7% subscribe. March, September, October and December all convert above 43%.

### Over-contacting reduces conversion

Clients called more than 3 to 4 times in this campaign are less likely to subscribe than those called once or twice.



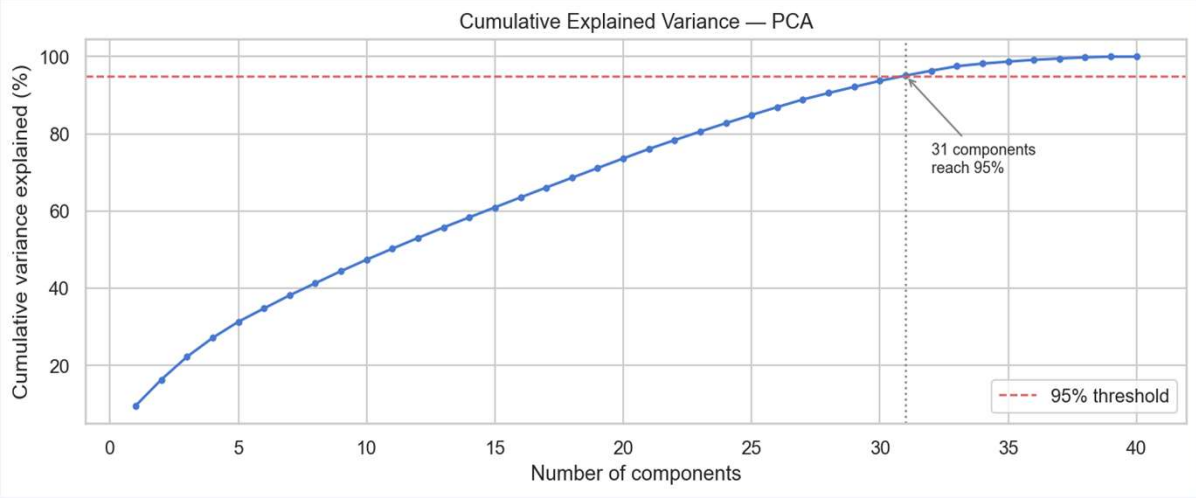
# PCA

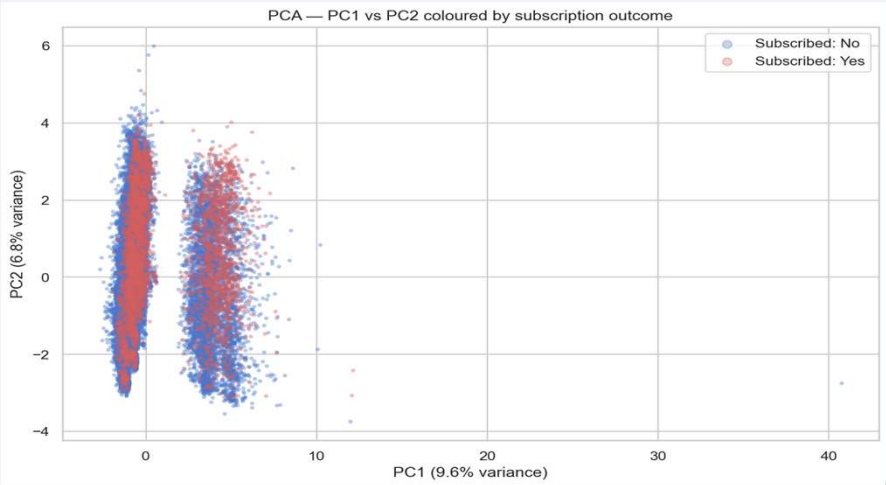
Dimensionality reduction: compressing 40 features into fewer dimensions for visualisation

PCA stands for Principal Component Analysis. It is a technique that takes a dataset with many features and finds a smaller set of directions that capture most of the variation.

Think of it like finding the best angle to look at a cloud of points so you can see as much spread as possible. We start with 40 features and PCA finds the directions where the data varies most.

We use it to ask one question: when we compress 40 features down to just two dimensions, can we see subscribers and non-subscribers forming separate groups? If yes, a simple model would work. If no, we need something more powerful.




| 40 | 31 | 16.4% |
|---|---|---|
| input features | components for 95% variance | variance in first 2 PCs |

# t-SNE

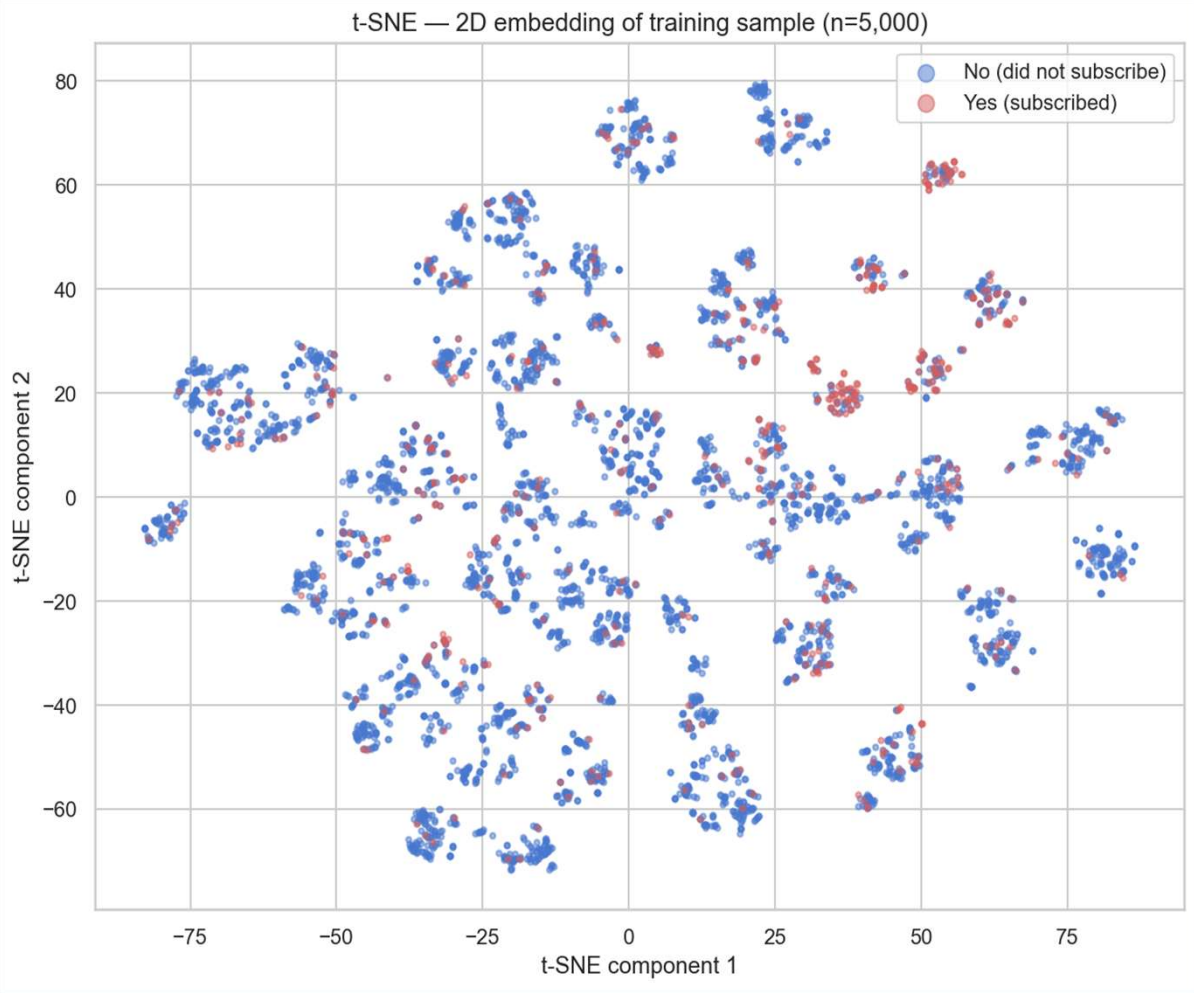
A second visualisation technique to check for hidden structure in the data

t-SNE is another dimensionality reduction technique, but it works differently to PCA. Instead of finding directions of maximum variance, it tries to keep similar points close together in 2D space.

It is particularly good at revealing clusters and structure that PCA might miss because it focuses on local similarity rather than global variance.

Because it is slow to compute on large datasets, we ran it on a random sample of 5,000 training rows. Each point is one client, coloured by whether they subscribed.

**The result confirms what PCA showed: the yes and no points are mixed throughout the plot rather than forming one clean cluster each. This rules out simple linear classifiers and confirms we need non-linear models.**



# Models

Three classifiers trained inside scikit-learn Pipelines with cross-validation

Each model is wrapped in a Pipeline: StandardScaler runs first, then the classifier. This is important because it means the scaler only ever sees training data during cross-validation and never leaks test set statistics into the model.

## Random Forest

**CV ROC-AUC: 0.7941**

30 random combinations via RandomizedSearchCV, 5-fold cross-validation

Builds hundreds of decision trees, each trained on a random subset of the data and features. Their predictions are averaged. This reduces overfitting and works well with imbalanced classes when class_weight=balanced is set.

## Gradient Boosted Trees

**CV ROC-AUC: 0.7972**

30 random combinations via RandomizedSearchCV, 5-fold cross-validation

Builds trees one at a time. Each new tree corrects the mistakes made by the previous ones. The model gradually focuses effort on the hardest-to-classify clients. Naturally handles imbalance through its error-correction process.

## SVM - RBF Kernel

**CV ROC-AUC: 0.7776**

Full grid of 12 combinations via GridSearchCV, 5-fold cross-validation

Finds the widest possible boundary between classes by transforming the data into a higher-dimensional space. The RBF kernel allows curved, non-linear boundaries. class_weight=balanced applied. Significantly slower to train than the tree models.

# Results

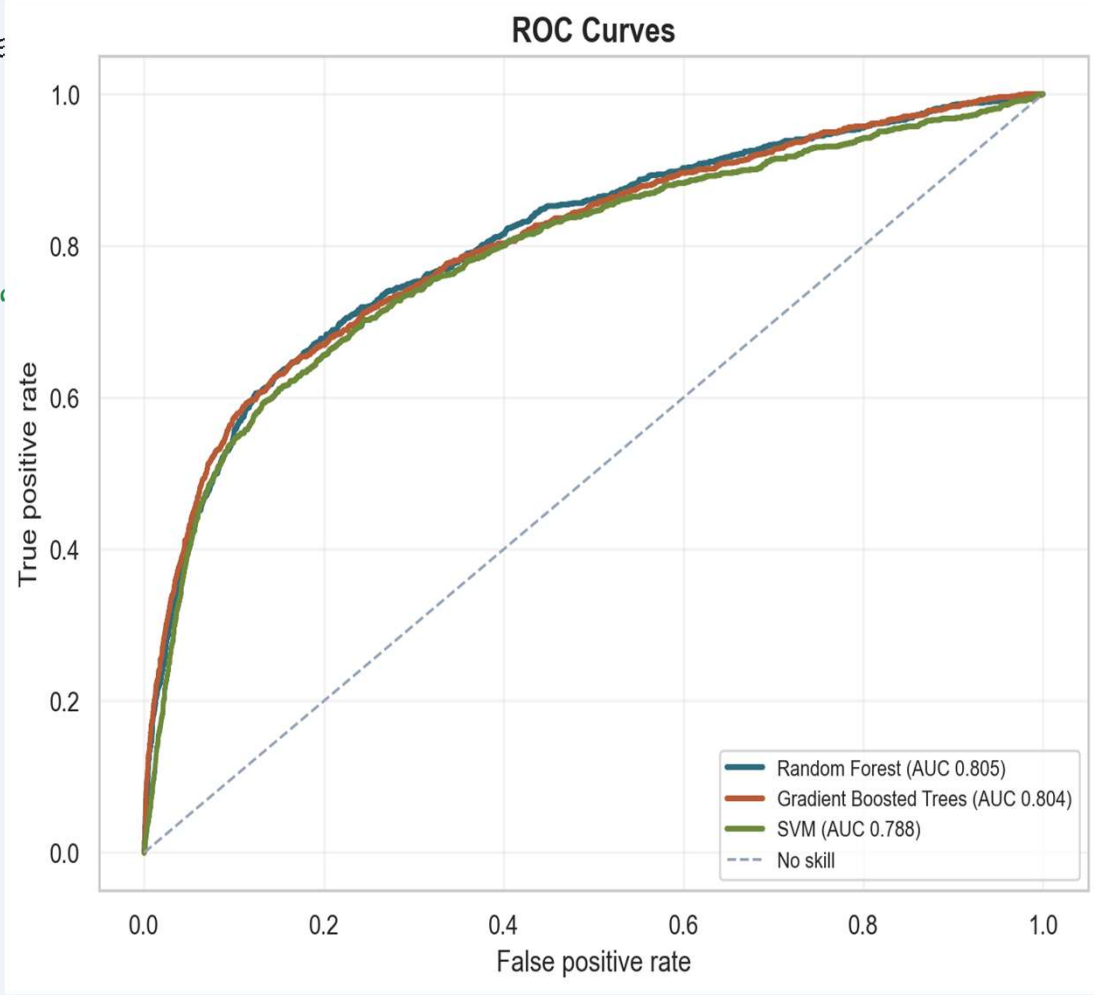
Evaluated on 9,043 test rows the models never saw during training

Accuracy alone is misleading here. A model predicting no every time
and F1 for class 1.

| Model | ROC-AUC | Recall - Yes | F1 - Yes | |
|---|---|---|---|---|
| **Random Forest** | 0.8054 | 0.539 | 0.473 | Recommended |
| GBT | 0.8036 | 0.266 | 0.375 | |
| SVM | 0.7881 | 0.591 | 0.458 | |

GBT has the highest accuracy but recall for subscribers of just 26.6%. It predicts no most of the time and misses nearly three in four potential subscribers.

**Random Forest finds 53.9% of subscribers with the best overall ranking quality.**



# Insight

Feature importances: what did the models actually learn?

Tree-based models like Random Forest and GBT can tell us which features were most useful for making predictions. This is measured by how much each feature reduces prediction error across all trees.

Both models agree on the top features, which gives us confidence in the result.

### poutcome - success

The strongest signal by a wide margin. Matches the 64.7% we saw in EDA.

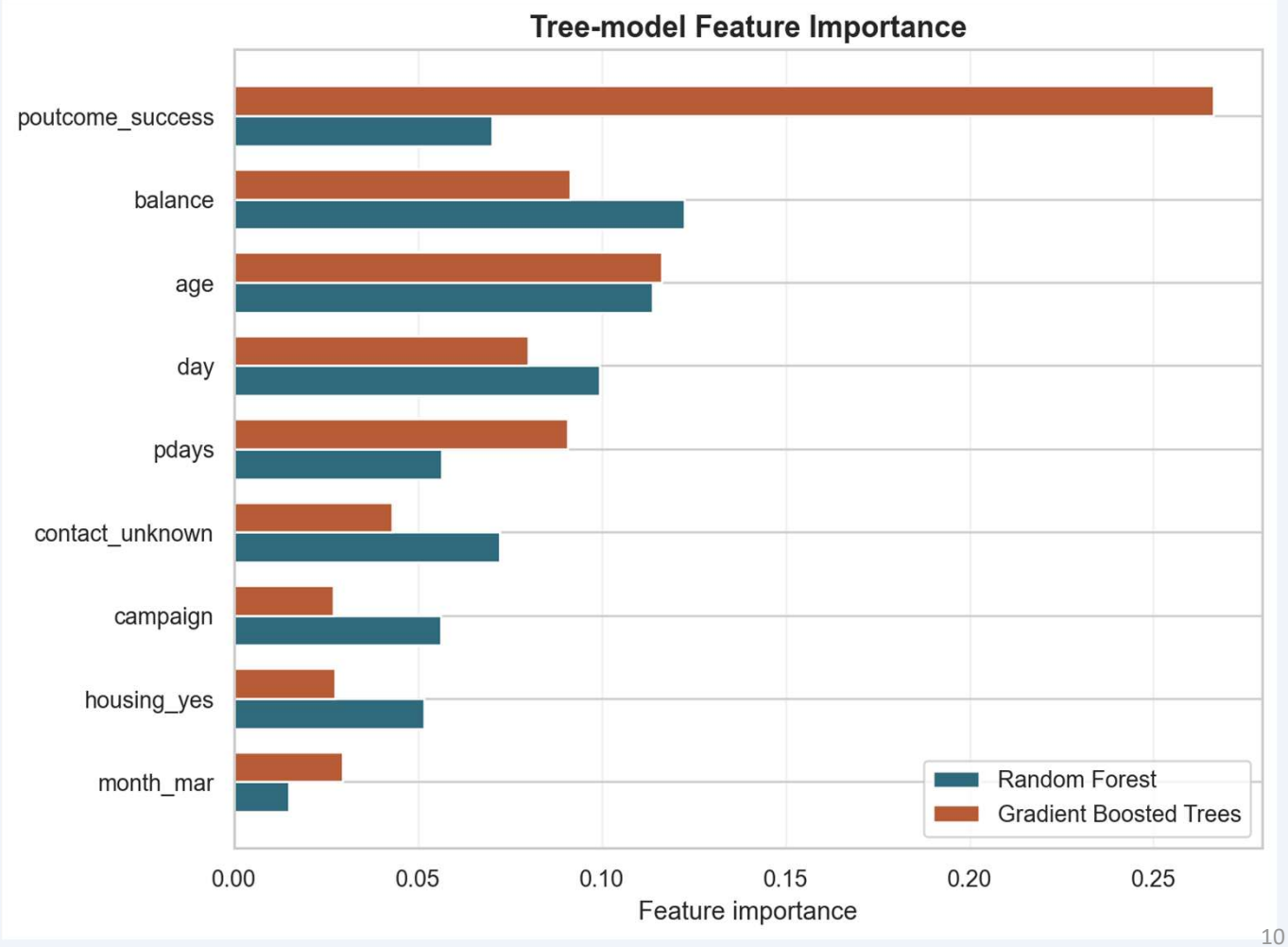
### balance and age

Wealthier and older clients are more likely to subscribe.

### contacted_before and pdays

The feature we engineered from the pdays sentinel appears in both models, justifying that preprocessing decision.

Note: SVM has no built-in feature importances because its decision boundary is defined by support vectors in a



# Conclusion

### Recommended Model

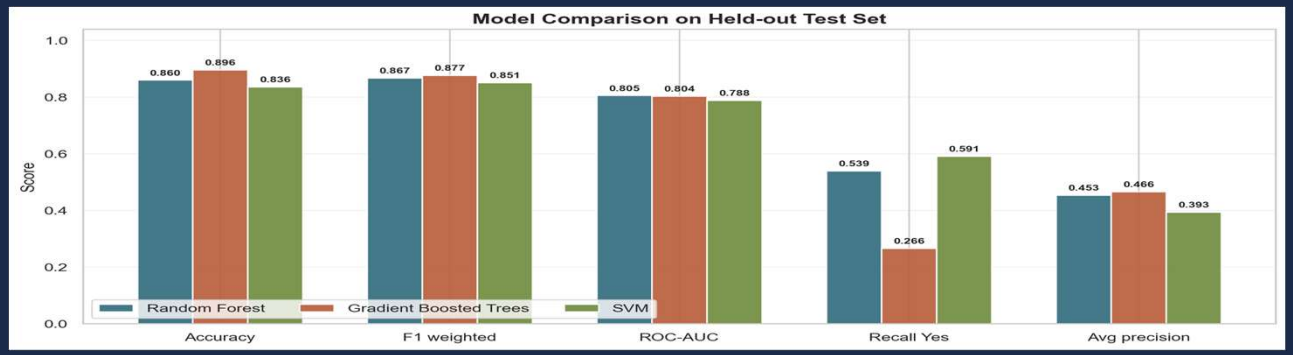
## Random Forest

Best ROC-AUC (0.8054) and best F1 for subscribers (0.473). Correctly identifies 53.9% of all clients who would subscribe, far ahead of GBT which finds only 26.6%.

GBT accuracy of 89.6% is misleading: it misses 73% of subscribers. For a bank campaign that is not a useful result.

### Key Takeaways

Prior campaign outcome is the strongest predictor by a large margin. The bank should prioritise re-contacting past successes.

Campaign timing matters more than call volume. Q4 months consistently outperform May despite lower volumes.

Over-contacting within a campaign reduces conversion. A contact limit per client would likely improve results.